

National Internet Exchange of India

(Empowering Netizens)



Internet Corporation For Assigned Names And Numbers

What If the Root Fails? Urging India to Build Its Own DNS Sovereignty

SANJAY K K
Internet Governance Intern *National Internet Exchange of India*
Student, K S Rangasamy College of Technology, Dept. of Electronics and Communication Engineering

Mentor: Dr. Govind
Former Senior Director, Department of Electronics and Information Technology, Government of India.
Ex-CEO, National Internet Exchange of India,Delhi

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ABSTRACT

The Domain Name System (DNS) operates through a hierarchical structure, with the root zone located at the apex, globally managed by the Internet Assigned Numbers Authority (IANA) under the auspices of ICANN. This root zone serves as the foundational point for resolving all domain names, including country-code top-level domains (ccTLDs) like .in. While India oversees the .in domain locally through NIXI and IN Registry, it is noteworthy that the root zone for .in is not situated within the nation; consequently, DNS queries must still be directed to the 13 globally governed root servers for resolution. Any disruption to the root zone, resulting from technical failures, cyberattacks, or geopolitical tensions, has the potential to render .in domains inaccessible. Countries such as Russia and China have addressed this challenge by establishing sovereign DNS systems, including replicas of the National Root Zone and legal governance. Currently, India does not possess a similar infrastructure. Although it is impractical for India to create its own root system, the question arises: why has India not developed a National Root Zone copy akin to those in Russia and China? This paper articulates the importance for India to construct its own DNS root zone copy infrastructure to achieve DNS sovereignty. By initiating this effort, India can ensure the uninterrupted functionality of the .in domain, even amid global Internet disruptions, network partitioning, or disconnection from the IANA-managed root system.

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INTRODUCTION

The Domain Name System (DNS) is often referred to as the "phonebook of the internet," translating human-readable domain names into IP addresses that computers use to communicate. At the top of this hierarchical system lies the DNS root zone, a critical component managed by the **Internet Assigned Numbers Authority (IANA)** under the oversight of the **Internet Corporation for Assigned Names and Numbers (ICANN).** The root zone is served by 13 logical root servers (labeled A to M), which are replicated across the world using Anycast to enhance redundancy and performance.

While this globally distributed architecture ensures stability, the core management and control of the DNS root remain centralized, primarily outside India. Despite hosting more than 40 mirror instances of root servers, India still depends on the global root zone for DNS resolution. These mirror servers (also known as Anycast instances) improve speed and resilience but are not authoritative and rely on updates from the central root servers.

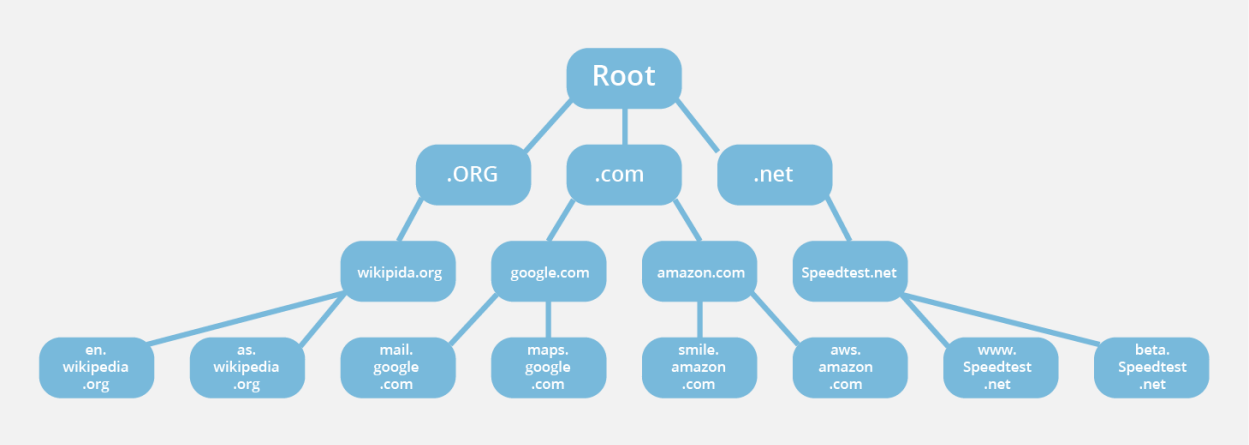
This centralization raises concerns about **resilience, sovereignty, and continuity** in the event of geopolitical disruptions, cyberattacks, or global internet fragmentation. In such scenarios, India's DNS infrastructure could experience delays or failures due to its reliance on external DNS root infrastructure.

This paper proposes exploring the feasibility of a **National Root Zone Copy** for India **a locally hosted and regularly synchronized version of the global root zone** to enhance the country's ability to resolve DNS queries even in the absence of international connectivity. Drawing on the experiences of countries like China and Russia, which have taken steps to build sovereign DNS capabilities, this study aims to identify a **balanced, policy-compliant approach** that supports resilience without undermining the global DNS trust model maintained by ICANN.

I. UNDERSTANDING THE DNS ARCHITECTURE

The Domain Name System (DNS) is the backbone of the Internet, allowing users to access online resources using easy-to-remember domain names instead of complex IP addresses. It is distributed database that translates human-readable domain names like "google.com" into numerical IP addresses that machines use to communicate.
Domains: - A **domain** is a readable, user-friendly name that maps to the numerical IP addresses of computers on the internet. It acts as the digital identity of a website and helps users access websites without needing to remember complex numbers
A domain name usually divided into two categories:

- **Generic Top-Level Domain (gTLD:**
  These are open for general use, not restricted by country like, **com, .edu, .org**
- **Country Code Top-Level Domains (ccTLDs):**
  These are reserved and managed by specific countries or territories, like
  - **.in - India**
  - **.uk - United Kingdom**
  - **.jp - Japan**

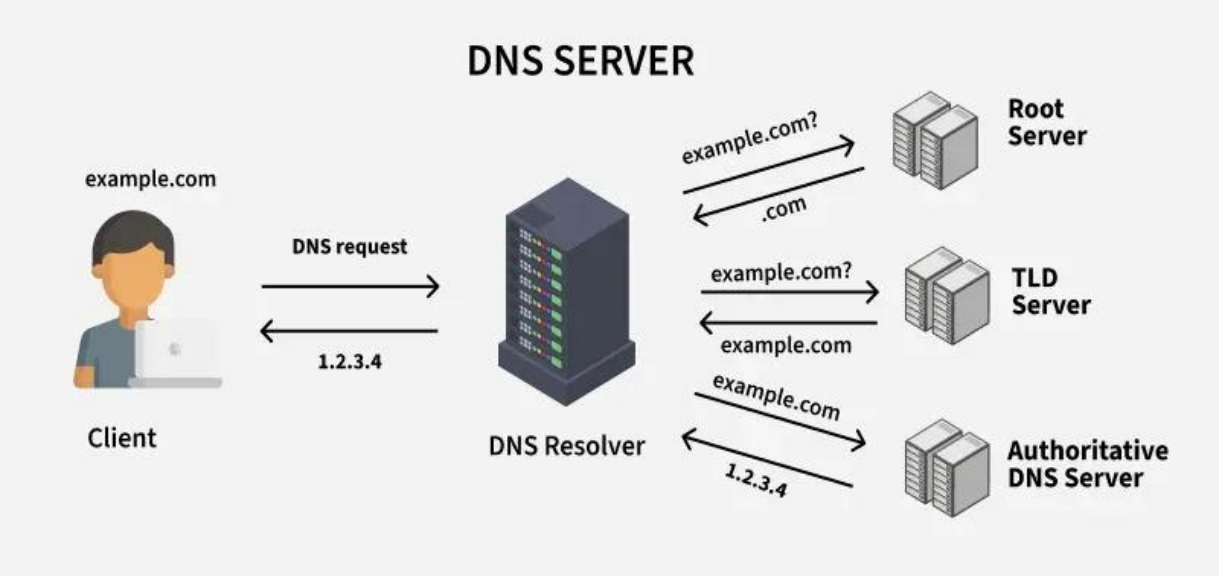


IP Address: An Internet Protocol (IP) address is a unique numerical identifier for every device or network that connects to the internet. Typically assigned by an internet service provider (**ISP**), an IP address is an online device address used for communicating across the Internet.
There are two versions of IP addresses that are commonly used on the internet**:** **IPv4** and **IPv6**. An IPv4 address is expressed as a set of **four dotted decimal numbers**, where each octet is separated by a period, such **as 192.168.35.4.**

The three digits in the first octet represent a particular network on the internet while the rest of the digits represent the actual host address within the local network, such as a workstation or a server. An IPv6 address represents eight groups of four **hexadecimal** digits separated by colons, such as **2620:cc:8000:1c82:544c:cc2e:f2fa:5a9b**.[1]

Conversion of Domain name into IP address:

When you type a website name like www.google.com into your browser, your computer doesn't actually understand names, it needs a number called an **IP address** to find and connect to that site. This is where the **Domain Name System (DNS)** steps in. Think of DNS like a phonebook for the Internet. Your computer first asks a **DNS resolver**, usually provided by your Internet service provider if it knows the IP address for the website. If it doesn't, it asks a special set of servers called **root servers**, which then point it to another set of servers called **TLD servers** (like for .com or .in). These servers then guide your computer to the exact place (called an **authoritative name server**) where the website's address is stored. Finally, your computer gets the IP address, connects to the site, and the webpage loads. All of this happens in the background within seconds so fast that most people don't even notice.



A. Root Server

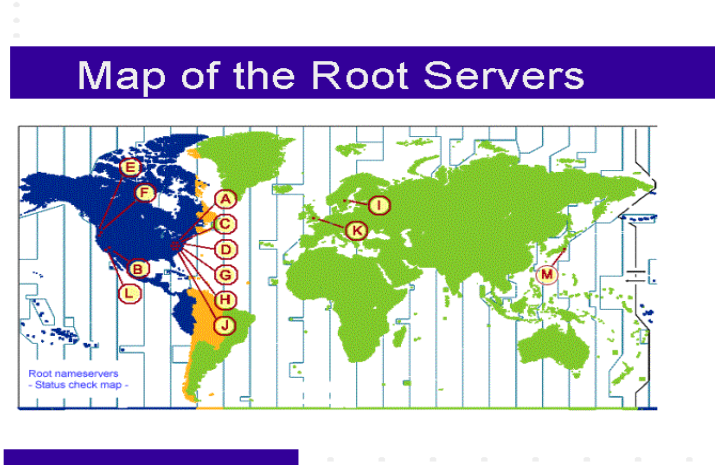
The **root zone** is like the master list at the top of the Domain Name System (DNS). It contains essential information that helps direct internet queries to the right **Top-Level Domain (TLD)** servers, such as those for `.com`, `.org`, or `.in`. Whenever a DNS lookup begins, the very first step is to refer to this root zone. The process of handling these first queries is known as the **root service**.

This root service is powered by a network of special servers called **root servers**, and the organizations that run them are called **Root Server Operators (RSOs)**.[2] There are 13 root servers identified by letters A through M, managed by 12 RSOs, and they are distributed across the globe. Each of these root servers doesn't just exist in one location, they have **multiple instances** thanks to a technology called **Anycast**, which helps speed up response time and increase reliability. Together, these servers form what's known as the **root server system**, the backbone of the internet's domain resolution process.

The List of Root servers in the Wolrd as been displayed below:

List of Root Servers

HOSTNAME	IP ADDRESSES	OPERATOR
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	Verisign, Inc.
b.root-servers.net	170.247.170.2, 2801:1b8:10::b	University of Southern California, Information Sciences Institute
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	Verisign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project



Even though there are 13 root the **A-root server**, managed by Verisign, plays a foundational role in distributing the root zone file. Under an agreement with ICANN and the U.S. Department of Commerce, Verisign operates the **hidden primary** that receives signed updates to the root zone directly from IANA

Verisign prepares the master root zone file, carefully checks it for errors, and then disseminates it to secondary root server operators (the other 12 letters). This systematic distribution ensures that every root server A through M serves the same authoritative data

In essence, while all root servers perform the same role in responding to DNS queries, the **A-root acts as the initial gatekeeper** the first to receive official updates and initiate synchronization across the global root server ecosystem.

- Once updates to the root zone are made and cryptographically signed, they are sent to **Verisign**.
- Verisign operates the **hidden primary server**, which is not publicly accessible but serves as the authoritative source for root zone data distribution.
- From Verisign's A-root, updates are propagated to all 12 other root server operators and their many global instances via the **root zone distribution system**.



B. TLD Server

Once a DNS query is directed from the root server, it reaches the TLD server corresponding to the domain extension, such as .com, .org, or.in. These servers don't contain the full IP details of a website but know which authoritative name server holds that information.

TLD servers act as intermediaries that guide the DNS resolver to the exact source that can provide the final IP address of the requested domain.

C. Authoritative DNS Server

An **Authoritative DNS Server** is a critical part of the Domain Name System that contains and serves the actual DNS records for a domain name. Unlike recursive resolvers or caching servers, which temporarily store data, an authoritative server is the original source of truth for domain-specific information. Once a DNS query has been directed through the root server and then the TLD (Top-Level Domain) server (e.g., .com, .in, .org), it is passed to the authoritative server for the specific domain.

II. ROOT INSTANCES

Root server instances are individual deployments of the 13 logical root servers that form the backbone of the Domain Name System (DNS). While the protocol allows for only 13 root server identifiers (A through M), each of these can have multiple physical instances deployed across the world using a method called **anycast routing**.

A. Anycast Routing

Anycast is a networking technique used to route data to the nearest or most optimal server in a group of servers sharing the same IP address. In the context of DNS, it allows multiple **root server instances** around the world to respond to DNS queries using the same IP address. The routing infrastructure automatically directs the query to the closest instance based on network conditions.

As of now, there are hundreds of root server instances globally, maintained by 12 independent Root Server Operators (RSOs). These instances are strategically located in over 150 countries to ensure geographic diversity and load balancing. The B-root server still exists only in its original location, Los Angeles, operated by the Information Sciences Institute. But the remaining twelve root servers exist in multiple locations around the world.[3]

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How the DNS Root System Stays Resilient: Jumping Between Root Servers

One of the most reliable parts of the Internet is something we don't often see the DNS root server system. There are 13 main root servers, named from A to M, and each one has many copies (called instances) located all over the world using something called *anycast*. These servers help your computer begin its search for any website by pointing it in the right direction.

But what happens if one of these root servers doesn't respond?

That's where the system shows how smart and resilient it really is:

- **Every DNS resolver knows all 13 root servers.**
  When your device starts a query, it usually picks one of the 13 root servers at random. Let's say it picks A-root first.
- **If A-root doesn't respond, the resolver doesn't wait long.**
  It quickly tries the next one—maybe B-root—and continues until it finds a server that replies.

- **It happens automatically.**
  This switching happens in the background, without you ever noticing. There's no manual effort or user involvement.
- **Even within a single root server, there are multiple locations.**
  For example, A-root might have servers in several cities around the world. If one location is down, the request is rerouted to another. Only if all A-root instances are unreachable does it move to B-root.
- **All root servers give the same answers.**
  No matter which root server your request reaches, the response will be the same because they all use the same **global root zone data maintained by IANA.**

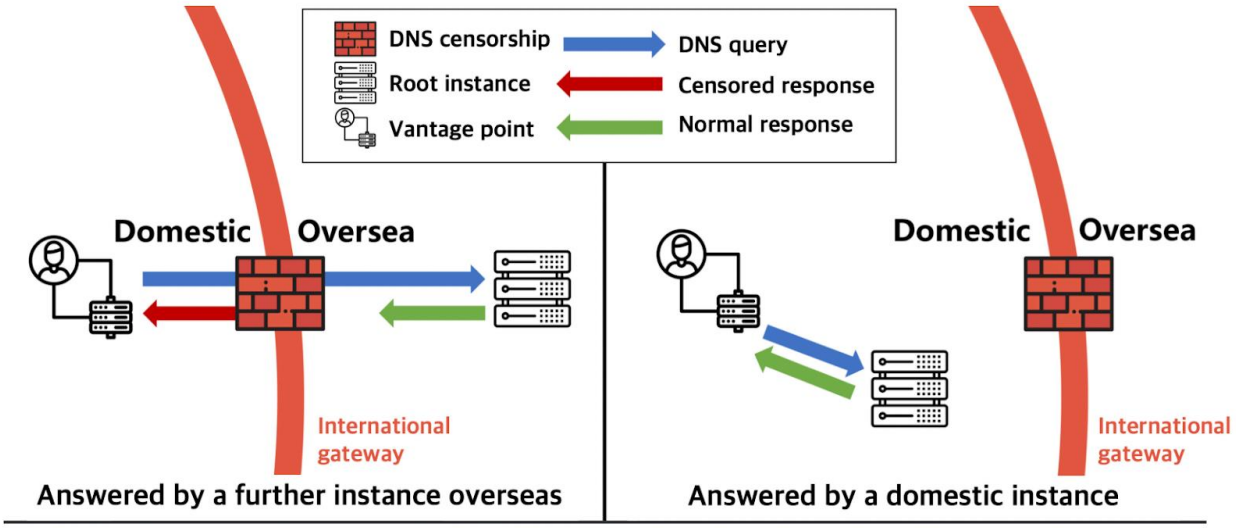
B. Root Instances in India

In India alone, there are more than **40 root server instances** hosted across major cities. Their presence strengthens India's Internet backbone by reducing dependency on international infrastructure and ensuring more efficient local DNS resolution.

Root Server Instances in India:

Root Server Type	City	Sponsor
J-root	Delhi	-
J-root	Mumbai	-
J-root	Gorakhpur	-
L-root	Mumbai	-
L-root	Kolkata	-
L-root	Mohali	-
L-root	Bangalore	-
L-root	Guwahati	-
I-root	Mumbai	NIXI
K-root	Delhi	NIXI
F-root	Chennai	NIXI
D-root	Mumbai	-
M-root	Mumbai	-

The National Internet Exchange of India (NIXI) has sponsored three root server instances, one I-root in Mumbai, one K-root at Delhi and an F-Root in Chennai, and further, there is also a D-root instance in Mumbai.[4]



Answered by an Overseas Instance:

The user's DNS query travels through an international gateway to reach root servers located outside the country. This path is slower, relies on foreign infrastructure, and may be subject to censorship or filtering before the response returns. It's vulnerable during geopolitical tensions or global network disruptions.

Answered by a Domestic Instance:

The DNS query is resolved within the country using a local root instance. This approach ensures faster resolution, improved reliability, and reduced dependence on international connectivity. It also enhances control over the DNS infrastructure, supporting national internet resilience.

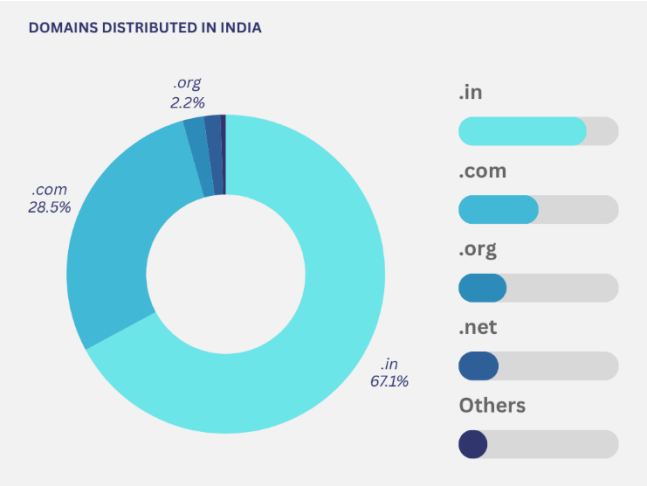
Though India already hosts several anycast root server instances, technically there is no difference between these and the original root servers. No matter where they are located, all root server instances perform the same function responding to DNS queries quickly and efficiently.

This brings us to the core question and problem statement of this paper: **Does India really need its own original root server or local root zone copy, or are the many anycast instances already present in the country sufficient for ensuring DNS resilience?**

III. .in DOMAIN

The **“.in” domain** is India’s **country code top-level domain (ccTLD)** and is an important part of the national Internet infrastructure. It is managed by the **National Internet Exchange of India (NIXI)** through its subsidiary, the **IN Registry**, under the delegation and policies of **ICANN and IANA**.

When a user types a website ending in “.in,” the DNS resolution process begins by contacting the **root servers**, then moves to the **TLD servers** responsible for the “.in” zone, and finally reaches the **authoritative name servers** that store the actual IP address of the domain. While root and TLD servers are globally distributed, **“.in” domains rely heavily on root zone access**, which is maintained externally.



Source: https://domainnamestat.com/statistics/country/IN

The chart shows that most websites in India use the **“.in” domain**, making up **67.1%** of all registered domains in the country. This means a large majority of people and organizations in India prefer to use a domain that reflects their national identity.
The **“.com” domain** comes next with **28.5%**, which is still significant but much lower than “.in”. Other domains like **“.org”** and **“.net”** are used very little, and the rest fall under “Others” with minimal usage. This clearly shows that **India heavily depends on the .in domain**, and if anything were to disrupt the DNS service related to it, it could affect a huge portion of the country's Internet activity.

A. .in Data in the Root Zone

Although the “.in” domain is managed and operated by NIXI within India, all the crucial data and delegation information for “.in” is still listed in the global root zone

file maintained by IANA under ICANN's oversight. This means that while day-to-day operations and management of the domain happen locally, the authoritative control and visibility of ".in" on the internet are rooted in the global DNS infrastructure. Every time a DNS query for a ".in" website is made, it ultimately depends on the global root zone to locate the authoritative servers. So, even with several Anycast root server instances hosted in India, the country still relies on an external, globally coordinated system

What happens if India's .in is cut off from the global DNS? Are we ready to keep our digital services running during such a disruption?

This is where the idea of DNS resilience comes into play

B. Internet Resilience

Internet resilience refers to the ability of a country's digital infrastructure to continue functioning smoothly even during unexpected disruptions be it due to cyberattacks, natural disasters, technical failures, or political shutdowns. In a globally connected system like the DNS, even a small issue at the top level can cascade into large-scale service interruptions. For a digitally advancing country like India, where everything from government portals to banking and education depends on the Internet, resilience isn't just a technical goal, it's a national priority.

C. Reasons for Network Breakdown

India's reliance on global DNS infrastructure opens up multiple points of failure that could disrupt access to the Internet during emergencies.

- **Submarine Cable Cuts**

  India's connectivity to global internet infrastructure heavily depends on undersea fibre optic cables. These submarine cables link India to global DNS servers, especially root server networks hosted overseas. If these cables are accidentally damaged such as by ship anchors or intentionally sabotaged, large

portions of the Internet could become inaccessible, including DNS resolution for even local domains like.**in.**

Understanding Submarine Cables

Submarine cables are the hidden backbone of the global Internet. These are fiber optic cables laid on the ocean floor, stretching thousands of kilometres to connect continents and countries. Although satellite communication exists, more than 95% of international data including web traffic, emails, videos, and DNS queries travels through submarine cables.

- Submarine cables are made of **optical fibers** bundled together and protected by layers of insulation and armor.
- They transmit data as light pulses, allowing **fast and high-capacity communication** between regions.
- Major cables connect India to Europe, the Middle East, Africa, and Southeast Asia such as **AAE-1**, **Tata TGN**, and **SEA-ME-WE** series.

Every time a user in India opens a global website or queries a domain like .com or even .in, the DNS resolution process may involve contacting a root server located outside India. These queries **travel via submarine cables** to global root server operators and IANA-managed infrastructure.

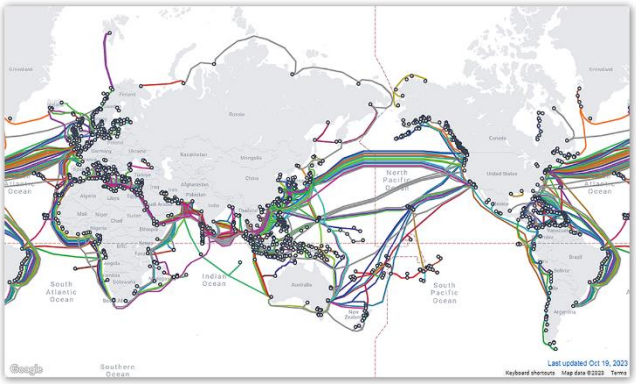


FIG: world's major submarine cable systems and landing stations.

- **Root Zone Unreachability**

All domain resolutions begin with the root zone. If the IANA-managed root zone becomes unreachable due to cyberattacks, misconfigurations, or intentional restrictions it can interrupt domain resolution at the very first step.

Even if India has 40+ root server instances, they mirror the global root zone and will be rendered ineffective if the primary source becomes inaccessible.

- **Cyber Warfare and DDoS Attacks**

  In the era of digital conflict, DNS infrastructure is often a target. A well-planned Distributed Denial of Service (DDoS) attack on root servers, TLD servers, or even national DNS infrastructure could paralyse domain name resolution. Such attacks can make even essential services like banking, healthcare, and government portals unavailable.

- **Misconfigurations or BGP Hijacks**

  The Border Gateway Protocol (BGP) governs how internet traffic is routed globally. A misconfigured or malicious BGP announcement can lead to DNS traffic being rerouted, dropped, or hijacked. This can cause widespread resolution failures and misdirection of user data.BGP hijacking is a type of cyberattack where attackers manipulate the Border Gateway Protocol (BGP) to divert internet traffic

D. Recent Causes of Network Partition and Disruption

- **Submarine Cable Cuts in the Red Sea (Feb–Mar 2024)**

  Multiple cables—including **Seacom**, **Tata AAE-1**, and **Europe India Gateway (EIG)** were damaged near Yemen's coast, reportedly by an abandoned ship's anchor amid conflict in the region.[5] These cuts forced global traffic to reroute, causing slower speeds and unreliable connections from Asia to Europe and Africa.

- **Bangladesh SEA-ME-WE 5 Cable Break (April 2024)**

  A break in Bangladesh's SEA-ME-WE 5 cable disrupted 1.7 Tbps of bandwidth, forcing traffic to be rerouted via India's land connections.[6] Internet speeds dropped for days, highlighting how even regional interruptions strain upstream DNS ecosystems.

- **June 25, 2016 Massive DDoS Attack on All 13 Root Servers**

On this date, a coordinated Distributed Denial of Service (DDoS) attack targeted the entire root server system, flooding each server with approximately 10 million packets per second (roughly 17 Gb/s). Some root servers experienced significant performance drops, with availability dipping to ~50% and resolution times spiking from ~3.7 Ms to 13 ms.[7] This event demonstrated that even highly distributed, anycast-enabled root servers are vulnerable to large-scale attacks.

- **May 2024 – 'C-Root' Server Glitch from Routing Misconfiguration**

  A misconfiguration related to peering within Cogent's network caused its 'C-Root' server to fall out of sync with the rest of the root server system for about four days, delaying root zone updates.[8]

- **2021–2024 – BGP Hijacking and Route Leaks**

  Numerous incidents including Vodafone Idea's 2021 leak of 30,000 BGP prefixes and various global route hijacks demonstrate persistent vulnerabilities. One flash example involved a hijack redirecting traffic to Amazon's DNS service, a clear reminder that DNS traffic can be stealthily rerouted via BGP manipulation.[9]

Countries approach towards DNS Resilience:

This refers to the ability of the Domain Name System (DNS) to continue functioning smoothly even in the face of failures such as cyberattacks, network outages, or physical infrastructure damage (like submarine cable cuts). A resilient DNS setup includes redundancy (multiple servers), geographic distribution (global and local instances), and caching mechanisms to minimize disruption.

The upcoming sections of this paper explore how countries like Russia and China have approached Internet resilience through sovereign DNS infrastructure, assess the technical foundations of DNS resolution and root zone dependencies, Each country has adopted its own approach to strengthening internet resilience based on its geopolitical priorities, technological capabilities, and policy frameworks.

IV. THE RUSSIAN MODEL OF ROOT ZONE RESILIENCE

In 2019, amidst rising concerns about cyber threats and geopolitical instability, Russia implemented a bold strategy to protect its national internet infrastructure. Through its **Sovereign Internet Law**, Russia created a **local copy of the global DNS root zone**, enabling the country to continue operating or accessing its Internet independently if it were ever cut off from the rest of the world intentionally or due to a crisis.

Under the 'Sovereign Internet' law, Operators will be required to install Roskomnadzor's TMCT on their systems and routinely provide detailed routing information to the regulator. They will also have to give Roskomnadzor remote access to the TMCT.[11]

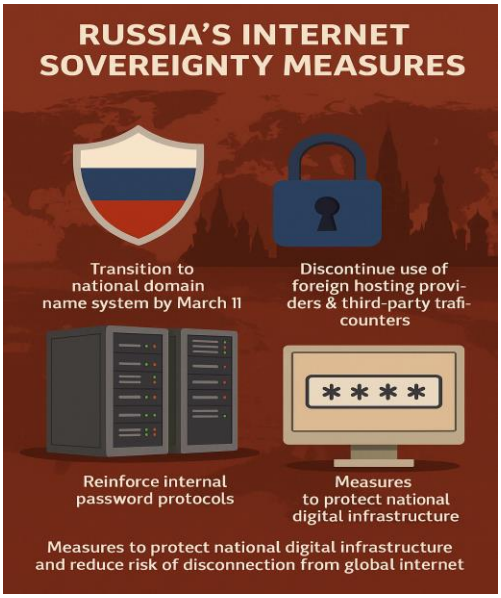
Key word: Technical Means of Countering Threats (TMCT)

RUNET:

Runet, short for Russian Internet, is Russia's state led national Internet infrastructure designed to operate independently from the global Internet.[10]

RU-CERT, operated under the Russian Institute for Public Networks an institution focused on developing computer networks for research, education, and core internet infrastructure in Russia serves as one of the country's key cybersecurity response bodies. As stated on its official platform, RU-CERT offers support to domestic and international legal entities, law enforcement agencies, and individuals in detecting, preventing, and mitigating cyber activities deemed illegal within Russian territory. It is also tasked with gathering and analyzing data on malware distribution and cyberattacks across Russia. RU-CERT represents Russia as a full member and official contact in the global Forum of Incident Response and Security Teams (FIRST) since 2001.

The Russian government mandated that all state-run websites and online services transition to the national domain name system by March 11. Additionally, it directed agencies to discontinue the use of foreign hosting providers and third-party traffic counters, while reinforcing internal password protocols. These measures aim to safeguard national digital infrastructure against potential cyberattacks and mitigate risks of disconnection from the global Internet.[11]



In a public address to the Russian Security Council, President Vladimir Putin emphasized the strategic importance of developing a sovereign internet infrastructure aimed at safeguarding the digital space of Russian users. He stated that this alternative system is designed to protect citizens from perceived manipulation and monopolization of internet services by Western nations. Furthermore, he proposed that Russia's independent infrastructure could potentially be extended to BRICS nations (Brazil, India, China, and South Africa) enhancing collective digital autonomy.

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However, the global response to this initiative has been mixed. Cybersecurity experts have expressed concerns that such a move could contribute to the **"Balkanization"** of the Internet, a term referring to the fragmentation of the global web into isolated, region-specific networks. Analysts from the UK's GCHQ have cautioned that Russia's alternative DNS system could be repurposed as a backup during cyber conflicts, or even leveraged for cyber offensives. Additionally, the creation of parallel DNS infrastructures may hinder international efforts to trace cybercriminal activity, misinformation campaigns, and state-sponsored threats online.[12]

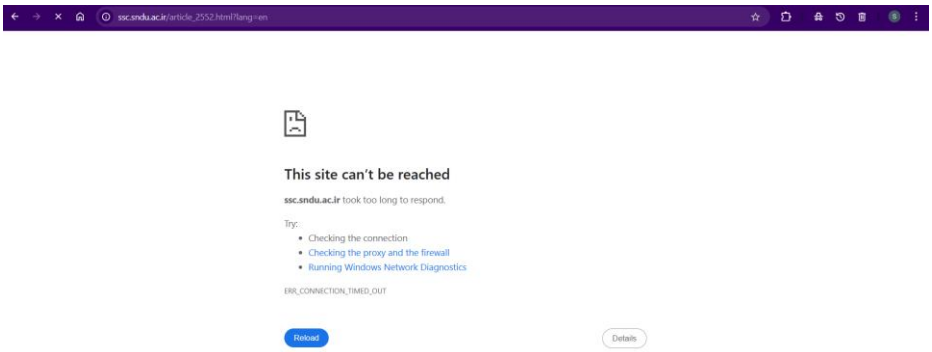
Note: *Balkanization in the internet context refers to dividing the global internet into smaller, region-specific segments, potentially leading to interoperability challenges and geopolitical tension.*

v. THE IRANIAN MODEL OF INTERNET RESILIENCE

Iran is widely recognized as one of the world's most Internet-blackout-prone countries. The government has repeatedly shut down the Internet during moments of political unrest or conflict, effectively isolating the country from the global web.

In mid-June 2025, during the height of tensions and Israeli airstrikes on Iran's infrastructure, the Iranian government enacted a near-total internet shutdown. Connectivity fell by approximately 97%, effectively isolating the country nearly entirely from the global Internet.[13]

- **Nationwide Internet Disruption:** Iran initiated a near-total internet shutdown, disconnecting large parts of the country from the global web.
- **Blocking of Major Platforms**: Popular services such as WhatsApp, Twitter (now X), Telegram, Instagram, and Facebook were completely inaccessible during the blackout, impacting communication and information flow.
- **Over 90% Drop in Connectivity**: Network monitoring agencies observed a steep decline in data traffic over 90% across several regions, confirming the scale of the disruption.
- **Effect on Daily Life and Economy**: Citizens were unable to access essential services, digital banking, or international news, while businesses suffered massive losses due to online inaccessibility.



The screenshot above shows an **"ERR_CONNECTION_TIMED_OUT"** message while trying to access an Iranian (.ir) domain **(ssc.sndu.ac.ir).**

Given the ongoing **Iran–Israel** conflict in June 2025, the inaccessibility of Iranian websites like this is most likely a result of state-imposed internet restrictions. According to recent reports, Iran has drastically limited international internet traffic and access to .ir domains from outside its borders as part of a broader cybersecurity defence mechanism during wartime. This includes:

- Blocking foreign IP access to reduce cyber threat exposure.
- Restricting outbound DNS requests to prevent information leakage.
- Routing national traffic internally through the National Information Network (NIN) for resilience and control.

So, the error message is not a random technical failure it's a deliberate access restriction implemented to shield Iran's digital infrastructure and communications during the conflict.

In response to recurring unrest and perceived external threats, Iran has developed the **National Information Network (NIN)** a state-controlled infrastructure designed to function as an alternative domestic Internet.

National Information Network (NIN):

NIN is built as a separate domestic network that can function even if Iran is cut off from the global internet. It gives the government the ability to keep basic services and internal websites running during external shutdowns. During international blackouts, NIN ensures that local banks, government portals, health systems, and educational platforms continue to operate without disruption. Many Iranian websites and platforms are hosted on domestic servers. This means they don't rely on international data routes, making them more resilient during outages or global DNS issues. NIN allows the Iranian government to block global platforms (like WhatsApp, Twitter, and Telegram) [14], while offering state-approved alternatives. This gives them more control during politically sensitive periods.

This approach strengthens Iran's cyber sovereignty and offers a level of resilience by providing a fallback network that continues to function during global disconnection or cyberattacks.

Operations of NIN:

NIN gives Iran digital sovereignty the ability to control what enters, exits, or is visible on its networks. It's a backup strategy against cyberattacks, sanctions, and information

warfare, making the country's Internet less dependent on international infrastructure. In times of protest or war (e.g., during the 2022 Mahsa Amini protests or the 2025 Iran–Israel tensions), the NIN allows the government to disconnect from the global internet while still maintaining critical internal digital operations.[15]

Iran has developed domestic versions of popular services **Soroush** instead of WhatsApp, **Aparat** instead of YouTube, **Bale** instead of Telegram that are hosted within the NIN.

Iran's digital network operates in two layers:

- Global Internet Layer: Users can access international websites when permitted.
- Domestic NIN Layer: When global access is shut off, users can still access Iranian websites, apps, and services (like email, news portals, banking, and government services).

DNS within NIN enables resolution of services like local banking (.bank.ir), e-learning platforms, and e-government sites even during complete global DNS isolation. This forms the core of the NIN's value maintaining operational continuity during hostile cyber or military scenarios.[16]



Aparat

Fig:



Fig: Soroush



Fig: Bale

VI. CHINA'S NATIONAL INTERNET SOVEREIGNTY MODEL

China presents a unique and advanced example of a nation-state prioritizing internet resilience through a deeply centralized and controlled digital infrastructure**. Cyber sovereignty** in the Chinese context refers to the belief that a country has the absolute right to govern and control the Internet within its own borders just as it would with its land, airspace, and waters. This concept is central to China's internet policy and national security strategy.[17]

Great Firewall of China:

The Golden Shield Project, widely known as the "Great Firewall of China," is a large-scale cybersecurity and censorship initiative overseen by the Ministry of Public Security of the Chinese government. Launched in 1998, the project is designed to monitor, filter, and control internet traffic within China. It employs a range of technical methods such as IP blocking to deny access to certain websites, packet filtering to scan data for restricted keywords, and DNS tampering to redirect or block domain name requests.[18] Additionally, the system integrates more advanced surveillance tools including facial recognition, speech analysis, and credit record monitoring. Over time, the project has become a cornerstone of China's approach to internet sovereignty. According to the OpenNet Initiative, China operates one of the most sophisticated internet censorship systems in the world, allowing it to tightly control online information and access for its citizens.



The evil side of Great Firewall is in addition to its conventional filtering methods, the Great Firewall of China employs advanced techniques to block online traffic. One such method is the Transmission Control Protocol (TCP) reset attack, where network connections to specific IP addresses or domains are deliberately disrupted by forcing both ends to drop the connection. Furthermore, the system is capable of conducting man-in-the-middle (MITM) attacks, particularly against encrypted connections using Transport Layer Security (TLS) or its predecessor, Secure Sockets Layer (SSL)[19]. Through these attacks, Chinese authorities can intercept and examine data that would otherwise be securely encrypted. By spoofing TLS certificates, the firewall effectively gains access to the content of encrypted traffic, allowing it to monitor or block **information that users assume to be private and protected**. This capability significantly enhances the state's control over internet communication and reinforces its broader cyber sovereignty strategy.

Services and websites blocked by the Great Firewall:
The People's Republic of China blocks an ever-growing list of services and websites with the Great Firewall.[20] The most notable services blocked include the following:

- BBC
- Discord
- Dropbox
- Facebook
- Netflix
- Instagram
- LinkedIn
- Reuters
- Twitter
- *The Economist*
- *The New York Times*
- *The Washington Post*
- Google search services
- Reddit
- YouTube
- Zoom
- Some Wikipedia entries

nixi Empowering Netizens

Cyber sovereignty: **China's Internet is so separated from the global Internet that it is more accurately described as a national Internet.**[21]
ChinaNet underpins China's vision of a "cyber-sovereign" Internet where the entire data lifecycle, from DNS resolution to content hosting, is handled domestically.
It forms the physical and logical core of this strategy.[22]

1.ChinaNet (国家公共互联网骨干网)

- One of the largest backbone networks in the world.
- Operated by state-owned telecom giants like **China Telecom**.
- Provides redundant and layered infrastructure across all provinces.

2. National DNS Infrastructure

- China maintains its own root name servers and a separate domain name resolution system.
- While still interoperable with the global DNS system, China can rely on its Internal DNS infrastructure if global root zones become unreachable.
- China also introduced localized TLDs in Chinese scripts e.g., .公司, .中国[23]

3. Unofficial / Alternative Root Servers

In parallel, China has developed its own root servers that serve a national root zone copy. These are not part of the global DNS root server system and can operate independently. These servers:

- Resolve Chinese-script TLDs (like .中国, .公司, .网络)
- Maintain their own zone data, controlled by Chinese authorities (e.g., Ministry of Industry and Information Technology – MIIT)
- Can be activated in emergency or isolated scenarios.

What Makes Them "Unauthorized"?

- They mirror the structure and function of the global root zone but are not synchronized or approved by IANA.They may serve alternative root zone data sometimes excluding or modifying records from the global root.This breaks the "single root" principle that underpins global DNS stability.[24]

Russia, Iran, and China have taken distinct paths to strengthen their internet resilience. **Russia developed Runet**, a sovereign Internet infrastructure with a national root zone copy. **Iran implemented the National Information Network (NIN)** to maintain access during blackouts. **China operates its own root servers and DNS system** under strict cyber sovereignty. While their methods vary, each country aims to ensure **continued connectivity and digital control during crises**, highlighting the strategic importance of national Internet resilience.

VII. INDIA'S DNS RESILIENCE

INTRODUCING THE NATIONAL ROOT ZONE COPY

"With nations like Russia, Iran, and China advancing their sovereign internet infrastructures, it is now up to India to chart its own path toward building a resilient DNS and network ecosystem that ensures uninterrupted connectivity during crises."

During the assessment of India's DNS infrastructure, a critical vulnerability became evident. Although the country hosts over 40 root server instances, they are all dependent on the global root zone managed outside national boundaries. In the event of a complete network partition caused by a submarine cable cut, cyberattack, or intentional disconnection India would lose access to root zone data, halting DNS resolution. This concern becomes even more serious when viewed in the context of national security. For instance, during recent war tensions with Pakistan, if such a network breakdown would have occurred, essential communication systems, including those used by the military and emergency services, could have failed. This dependency on external DNS roots poses a real and pressing threat to India's digital resilience during critical times.

In September 2015, India made a formal approach to the United States requesting that a root zone server be established within its territory. While the request was aligned with India's vision of digital sovereignty, the limitation imposed by the global cap of 13 logical root servers made it a technically and administratively complex proposal to realize.[25]

First, we had to understand why the number of root servers has remained fixed **at thirteen**, we need to look at how data is handled during DNS communication. When a user's computer queries the DNS, the server responses must fit within a single data packet for quick and efficient exchange. With IPv4, each of these packets can carry only up to **512** bytes. Since every root server's address takes up **32 bytes**, 13 servers together consume 416 bytes leaving just enough room (around 96 bytes) for the necessary protocol details.[26]

A commonly raised question in policy and technical discussions is: If India already hosts multiple mirror instances of global root servers, what is the need for any further change?

A practical solution identified to address potential national-level network disruptions is the establishment of a **National root zone copy**. While India currently relies on globally distributed root servers and multiple anycast instances, these are still dependent on international infrastructure and IANA-controlled root zone data.

A.ICP – 3 Policy:

For the root zone, ICANN issued the ICP-3 policy on July 9, 2001, which states that ***"to remain effective, the DNS requires a globally unique public namespace, and this is only possible if all domain names are assigned from a single, authoritative root."***[27]

As per the ICP-3 policy, all 13 root servers must operate under a **single, globally coordinated root zone.** This policy clearly states that there should be only one authoritative root zone for the Internet, managed exclusively by IANA under ICANN's oversight. No other root zone should run in parallel with IANA's, as doing so could lead to confusion, technical errors, and fragmentation of the Internet's global structure.

A. NATIONAL ROOT ZONE COPY

A National Root Zone Copy is a locally hosted and regularly synchronized replica of the root zone file maintained by IANA under ICANN. Unlike root server mirror instances, which are mere extensions of globally operated root servers, a root zone copy is a complete local storage of the root zone data itself not just a routing mechanism, but a critical component that allows DNS resolution data to be stored inside the country itself like the Root Zone.

How Does a National Root Zone Copy work?

We've already seen that **mirror instances** are copies of root servers basically, they help direct DNS queries faster by being closer to the users, but they still rely on the **main root zone**, which is managed globally by IANA under ICANN.

Instead of just copying a server, it creates a **full copy of the root zone itself**, the actual file that contains all the essential information about where to find top-level domains like **.com, .org, .in**, and so on.

- **Similar to Global Root, But Independent**

  Functionally, it behaves exactly like the global root zone. It gives out the same answers, follows the same protocol, and ensures that domain lookups still work. The only difference is: the data is stored and served from within the country, increasing Internet resilience and reducing dependence on external networks.

- **Boosts Resilience**

  This local setup ensures that even if there's a disruption in international connectivity like a **submarine cable cut, a cyberattack, or a wartime network partition** DNS resolution can continue uninterrupted inside the country. Critical services like **military communication, emergency services, and governance platforms** remain accessible.

- **Sovereignty and Control**

  Having an independent copy of the root zone puts greater control in national hands, aligning with digital sovereignty goals. It allows the country to manage and monitor DNS infrastructure independently, while still staying in sync with global standards.

NOTE: As per ICANN's ICP-3 policy, only a single authoritative root zone managed globally by IANA is permitted to operate on the public Internet. In alignment with this policy, the proposed National Root Zone Copy is not intended to replace or run in parallel with the global root. Instead, it is strictly designed as a **resilience mechanism**, to be activated only during critical disruptions (such as network partition, cyberattacks, or geopolitical crises). This ensures that DNS resolution can continue within the country while fully respecting the global DNS trust model and avoiding any policy violations.

CONCLUSION

The Internet we use every day runs on something most people never think about—the Domain Name System, or DNS. It's what turns web addresses like *google.com* into the IP numbers that computers understand. This whole process relies on a structured, global system where the root zone is the topmost layer, managed by IANA under ICANN. Even our very own *.in* domain, while managed by NIXI, ultimately connects back to this global root.

India does have over 40 root server instances, and they do help speed up access and reduce load times. But these are not independent—they're copies, not originals. They still depend on the master root zone that lives outside our country. If a global crisis occurred say a cyberattack, war, or even a major undersea cable cut we could suddenly find ourselves digitally stranded, with critical systems unable to function.

This paper explored how other countries have dealt with that risk. Russia built its own DNS backup system under Runet. China built a parallel DNS to stay in control of its internet. Iran created a national network that works even during international blackouts. Each country did it in its own way, but all with the same goal to make sure their internet keeps running no matter what.

This research proposes a clear, practical idea **a National Root Zone Copy for India**. Not a separate internet, not a political move just a technical safety net. It would act as a backup, used only in emergencies, and would still follow all the global rules set by ICANN. Think of it like a spare tire there if you need it, but not meant to replace the real thing.

With India now having the second-largest number of Internet users in the world, To truly become digitally independent, we need to go beyond access and speed we need resilience. And this idea, when built responsibly and collaboratively, could be a major step toward securing India's digital future.

"A National Root Zone Copy is not a requirement, but an essential resilience measure for India's secure digital future."

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